

AI-Powered Medical Chatbot for Heart Disease Detection Using Hugging Face

Project Report

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Abstract

Artificial Intelligence (AI) is transforming the healthcare sector by providing intelligent solutions for disease prediction and patient assistance. This project presents an AI-Powered Medical Chatbot for Heart Disease Detection that combines image classification and chatbot technology. The system accepts heart-related medical images, analyzes them using a pre-trained Hugging Face Vision Transformer (ViT) model, and generates prediction results with healthcare recommendations. A Gradio-based web interface provides a simple and interactive user experience. This project demonstrates the practical application of AI, deep learning, and natural language processing concepts in healthcare.

1. Introduction

Heart disease is one of the leading causes of death worldwide. Early diagnosis and proper medical consultation can help reduce health risks and improve patient outcomes. Traditional diagnosis often requires expert medical analysis and can be time-consuming.

Artificial Intelligence offers opportunities to assist healthcare professionals by providing preliminary assessments and automated insights. This project aims to develop an AI-powered chatbot capable of analyzing medical images and providing heart disease-related predictions and recommendations.

2. Objectives

The main objectives of this project are:

- To develop an AI-powered healthcare assistant.
- To accept and process heart-related medical images.
- To perform image classification using a Hugging Face pre-trained model.
- To provide prediction results and confidence scores.
- To generate healthcare recommendations through a chatbot interface.
- To demonstrate the practical use of AI in healthcare applications.

3. Technologies Used

Programming Language

- Python

Libraries and Frameworks

- Transformers
- PyTorch
- Gradio
- Pillow (PIL)

AI Model

- Google Vision Transformer (ViT)
- Model: google/vit-base-patch16-224

Development Environment

- Google Colab

4. System Architecture

The workflow of the system is as follows:

1. User uploads a medical image.
2. The image is processed by the application.
3. The Vision Transformer model analyzes the image.
4. Prediction confidence is calculated.
5. The chatbot generates a disease explanation.
6. Results are displayed through the Gradio interface.

5. Methodology

Image Upload

The user uploads a medical image through the Gradio interface.

Image Processing

The uploaded image is converted into a suitable format for AI analysis.

Prediction

The Hugging Face Vision Transformer model performs image classification and generates prediction scores.

Chatbot Response

Based on the prediction confidence score, the chatbot provides recommendations and healthcare guidance.

User Interface

The Gradio interface allows users to interact with the system easily through a web browser.

6. Implementation

The project was implemented using Google Colab. Required libraries were installed and imported successfully. A Hugging Face image-classification pipeline was loaded using the Vision Transformer model.

The system consists of three major components:

- AI Model Loading
- Disease Explanation Function
- Prediction Function and Gradio Interface

The chatbot displays prediction results, confidence scores, recommendations, and a disclaimer indicating that the application is intended for educational purposes only.

7. Results and Testing

The application was tested successfully using a sample chest X-ray image.

Test Case

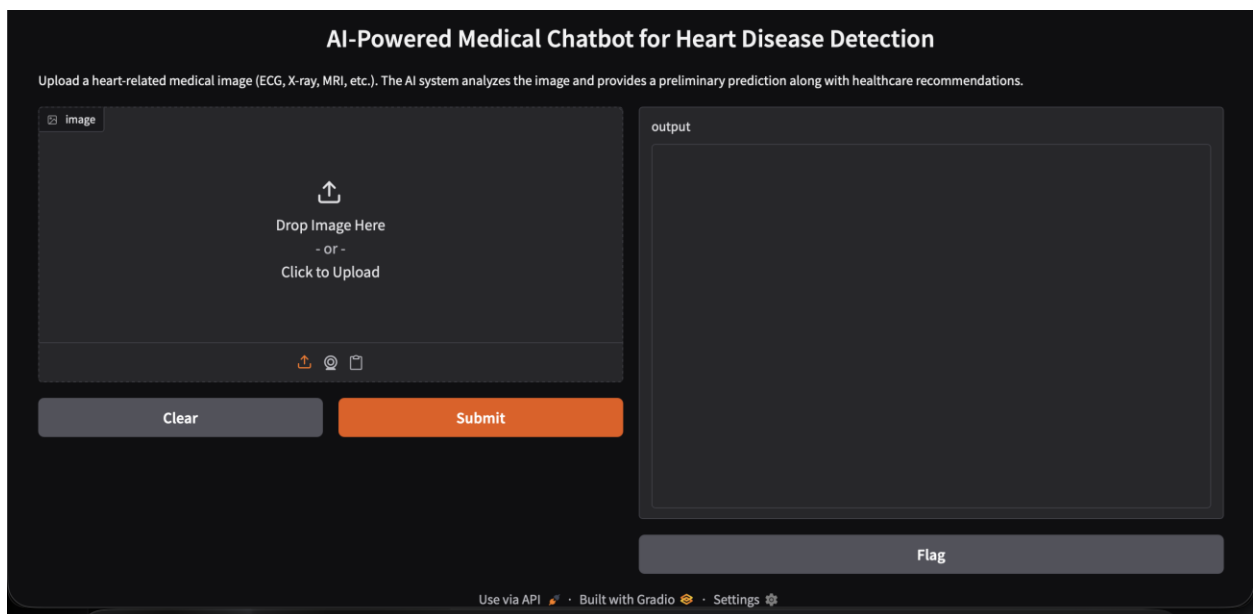
Input:

- Chest X-ray image uploaded through the Gradio interface.

Output:

- Prediction Result: NORMAL
- Confidence Score: 1.2%
- Recommendations:
 - Maintain a healthy diet
 - Exercise regularly
 - Schedule routine health checkups
 - Stay hydrated and manage stress

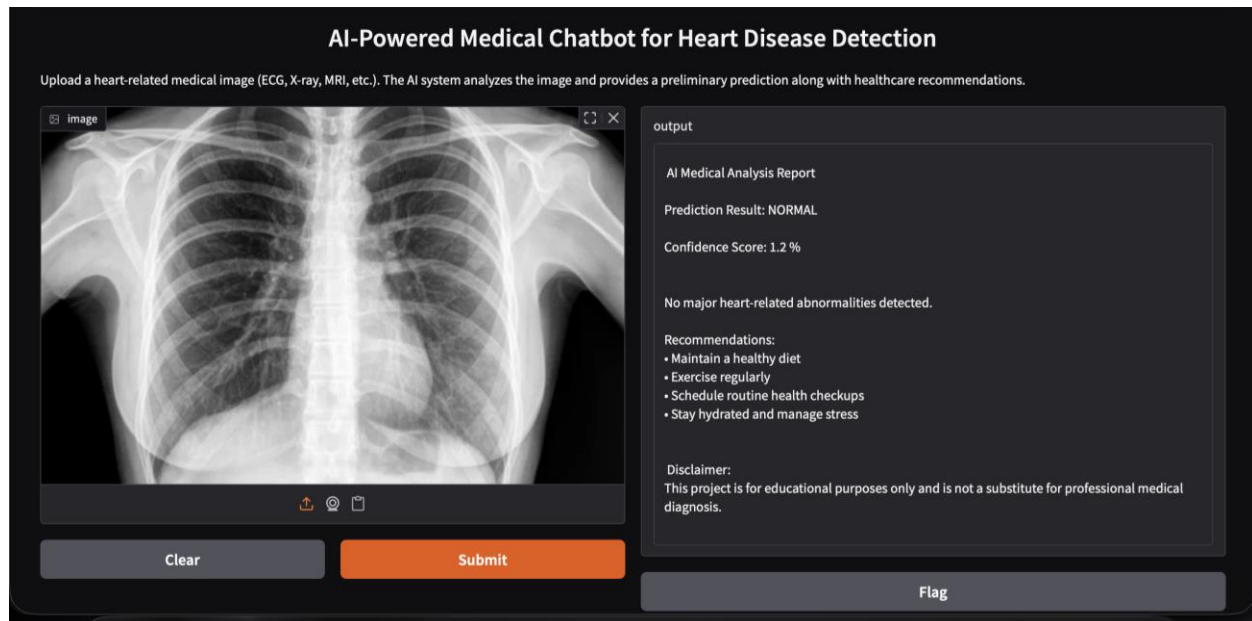
Figure 1: Application Home Screen



Description:

The home screen of the AI-Powered Medical Chatbot where users can upload heart-related medical images for analysis.

Figure 2: Prediction Result Screen



Description:

The application successfully analyzed the uploaded chest X-ray image and generated a prediction result. The AI model classified the image as NORMAL and displayed healthcare recommendations along with a disclaimer.

Observation

The system successfully performed image classification and generated a structured medical analysis report. The Gradio interface displayed results correctly, demonstrating successful integration of AI image classification and chatbot-based recommendations.

8. Advantages

- Easy-to-use interface.
- Fast preliminary assessment.
- Demonstrates practical AI integration.
- Uses pre-trained models without extensive training.
- Accessible through a web interface.

9. Limitations

- Uses a general pre-trained image classification model.
- Not intended for actual medical diagnosis.
- Requires internet connectivity for model downloads.
- Limited medical accuracy without specialized datasets.

10. Future Scope

Future enhancements may include:

- Training on specialized medical datasets.
- Integration with BioGPT and BioBERT models.
- PDF report generation.
- Patient and doctor login systems.
- Database integration using MongoDB or SQL.
- React.js frontend and FastAPI backend.
- Voice-enabled chatbot support.

11. Conclusion

The AI-Powered Medical Chatbot for Heart Disease Detection successfully demonstrates the application of Artificial Intelligence in healthcare. The project integrates image classification, chatbot technology, and a user-friendly interface to provide preliminary health-related insights. The successful implementation of the system highlights the potential of AI-assisted healthcare solutions and serves as a foundation for future improvements and real-world medical applications.